Novel Ferroelectric Tunnel FET-based Computing-in-memory with In-situ XOR Cipher-Encrypted AND-Type Multiply-Accumulate for Secure Edge AI

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Abstract

In this work, a novel ferroelectric tunnel FET (FeTFET) based computing-in-memory (CIM) system incorporating in-situ XOR cipher-encrypted AND-type multiply-accumulate (MAC) functionality is proposed and experimentally demonstrated for the first time. Leveraging the dual-modulation effect and ambipolarity of FeTFET, the activation (x) is represented by drain-controlled band-to-band tunneling (BTBT), while XOR operations between the Key and stored encrypted weight (w_e) are performed by ferroelectric-gate-controlled non-monotonic BTBT at both drain and source junctions. The in-situ XOR cipher-encrypted AND-type multiplication is realized within a single FeTFET, fabricated on a 14 nm FinFET platform, thereby eliminating decryption circuits. Based on the design, encrypted neural networks are demonstrated with reduced hardware costs and high energy efficiency, highlighting its potential for secure edge AI applications.
(*Keywords*: Ferroelectric tunnel FET, computing-in-memory, secure AI, XOR cipher)

Introduction

Computing in memory (CIM) architectures based on high-density non-volatile memory (NVM) have garnered significant attention [1, 2]. The raw weight matrices stored in CIM are susceptible to security breaches, presenting potential vulnerabilities (Fig. 1a) [3]. Encrypting CIM with an authorized Key is essential for secure edge AI. Recently, several secure CIM schemes employing in-situ XOR cipher in parallel expandable AND-type multiply-accumulate (MAC) computations have been explored [4, 5], where the Key and activation (x) of neural networks are applied to the CIM array concurrently (Fig. 1b). However, the encrypted weight (w_e) cell designs tend to be complicated, requiring complementary encrypted weight (w_e) storage and Key inputs for MAC with in-situ decryption, such as 2T-2RRAM configurations [5] or the SRAM with split word-line (WL) [4], suffering from high hardware cost and energy consumption. In our previous work [6], by algorithmically combining XNOR-type MAC with XOR cipher, and then utilizing gate-controlled ambipolar BTBT mechanism, we proposed a ferroelectric tunnel FET (FeTFET) based non-volatile CIM (nvCIM) array, which features single-transistor w_e along with row-wise XOR gates for decryption (Fig. 2a).

In this work, we further propose a AND-type MAC with in-situ XOR cipher, enhancing precision expansion capability while maintaining a lower cumulative current compared with our previously proposed XNOR-type encrypted MAC (eMAC). By simultaneously leveraging the drain- and gate-controlled BTBT, AND-type eMAC is experimentally demonstrated within a single FeTFET device, eliminating extra decryption gates (Fig. 2b). Based on this design, encrypted neural networks for pattern recognition are demonstrated for secure edge AI with reduced hardware cost and high energy efficiency.

Design of FeTFET-based nvCIM with in-situ XOR cipher-encrypted AND-type MAC

Fig. 2b shows the proposed FeTFET-based encrypted nvCIM array design. The XOR cipher, as an effective cryptography method for neural networks, is used for row-wise weight encryption (Fig. 2c), and the Key is applied to the word-line (WL) shared by the gates of each row's FeTFET. The AND-type multiplication with expandable precision is employed for MAC operation, and the activation (x) is applied to the read-word-line (RWL) shared by the drain of each row.

XOR cipher-encrypted AND-type MAC: As shown in Fig. 2b, the correct partial sum of MAC is obtained by two steps. First, XOR operation is applied to the ciphertext weight ($w_{e_i,j}$) and the authorized Key to retrieve the plaintext weight ($w_{i,j}$). Second, AND operation is performed between $w_{e_i,j}$ and x_i for multiplication. Finally, the bitwise multiplication results are accumulated along source-lines (SLs) as current. Both XOR cipher and AND-type multiplication can be simultaneously implemented by a single three-terminal FeTFET device with ambipolarity. Thus, by advancing the algorithm, the row-wise XOR gates presented in our earlier work [6] can be effectively eliminated.

FeTFET-based AND operation with XOR cipher: The FeTFET structure integrates a ferroelectric (FE) layer into the gate of the tunnel FET (TFET) with gate-controlled reverse-biased P-I-N structure (Fig. 2d). When a relatively large write voltage is applied, the w_e can be stored in FeTFET with different non-volatile FE polarization states, modulating the threshold voltage (V_{TH}), and shifting the ambipolar I_D-V_G curves. The x_i of "0/1" is applied through the drain voltage (V_D) of 0V/V_{DD} in RWLs. The Key bit of "0/1" is set through the readout gate voltage ($V_{G\text{-}read}$) of V_L/V_H, allowing the FeTFET to operate in a non-destructive read mode. For $x_i = 0$ (V_D=0V), the device generally exhibits a negligible low current. For $x_i = 1$ (V_D=VDD), only when the key is correct can FeTFET displays a drain current (I_D) corresponding to the

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plaintext weight (w) of "0/1", indicating the XOR cipher-encrypted AND operation for multiplication.

Therefore, this approach results in a higher sparsity of MAC outcomes, which is advantageous for reducing energy consumption compared with our previous proposed XNOR-based MAC scheme [6]. Moreover, this design eliminates the need of row-wise XOR gates for decryption [6], as well as the complementary weight storage and key-input in conventional encrypted CIM schemes [4, 5].

FeTFET-based XOR-encrypted AND-type MAC

A. Characteristics of FeTFET for in-situ XOR cipher

The proposed FeTFET is experimentally constructed by connecting an FE layer to the gate of TFET. The tungsten top electrode is paired with a 10nm $Hf_{0.5}Zr_{0.5}O_2$ FE layer to optimize write- and read-disturb immunity of FeTFET array (Fig. 3a-c) [7]. The TFET structure is fabricated based on a 14nm FinFET platform, which exhibits distinct ambipolar BTBT current (Fig. 3d) due to a specialized layout without a drain underlap region. Because of the unique dual-modulation effects [8], the BTBT windows and ambipolar current are controlled by both gate and drain voltage, representing the Key and x inputs, respectively. The I_D-V_G characteristics of FeTFET device exhibits ambipolarity with typical FE hysteresis under relatively high sweep voltage (Fig. 3e), due to the FE polarization switching, indicating the w_e storage capability.

B. FeTFET-based AND-type MAC with XOR cipher

Based on the fabricated FeTFET, the XOR cipher-encrypted AND-type multiplication behavior is experimentally characterized (Fig. 4). During initialization, the encrypted weight w_e, dependent on both w and the authorized key (Fig. 4a), is programed into FeTFET with different V_{TH} states and BTBT location. During the implicitly decrypted multiplication phase, the input x_i (V_X) and key (V_{key}) are respectively applied to the drain and gate of FeTFET. When $x_i = 0$, the I_D maintains the off state (I_L). When $x_i = 1$ (Fig. 4b-d), the correct product ($x \times w$) is only represented by different drain currents (I_L/I_H) when the authorized key is applied. If an unauthorized key is used, the product results are incorrect, leading to mistake in the MAC computation, thus achieving security enhanced nvCIM. Compared with our previous XNOR-eMAC, the AND-type eMAC eliminates the decryption gates by leveraging the drain terminal.

FeTFET-based encrypted nvCIM macro

Based on the proposed FeTFET-based nvCIM with XOR cipher-encrypted AND-type MAC, encrypted neural networks are demonstrated by utilizing our developed FeTFET model, featuring ambipolar BTBT in both source and drain junctions [6] (Fig. 5). The precision of x and w in nvCIM is extended through serial bits and multiple cells (Fig. 6a). If the key is incorrect, the accumulated current (I_{SL}) of each column shows errors (Fig. 6b), directly resulting in a significant accuracy degradation when using an unauthorized key (Fig. 7a). Moreover, the energy efficiency of FeTFET-based AND-type encrypted MAC (eMAC) is evaluated (Fig. 7b), showing improvement over SRAM-based scheme and further improvement compared with our previous FeTFET-based XNOR-type eMAC.

Conclusion

This work reports a novel FeTFET-based nvCIM with in-situ XOR cipher, and experimentally demonstrates precision-expandable AND-type MAC with implicitly decryption within a single-transistor w_e cell, enabling significantly hardware cost reduction (Table I). Based on our proposed design, the encrypted neural networks are implemented with high energy efficiency, highlighting its great potential for secure edge AI applications.

Acknowledgments

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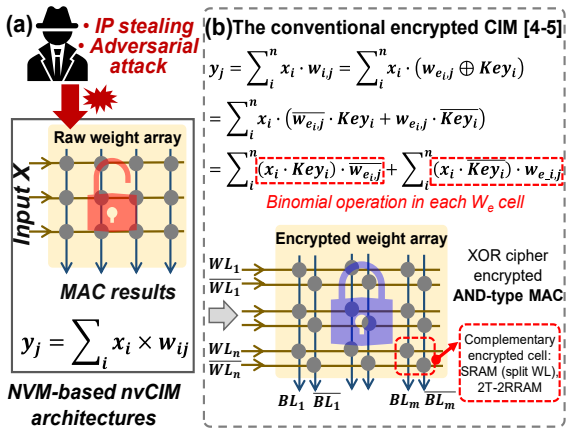


Fig. 1 (a) The security vulnerabilities of non-volatile memory (NVM) based computing-in-memory (CIM) architectures. (b) The conventional encrypted CIM schemes require complementary form for encrypted weight (w_e) storage and the Key inputs.

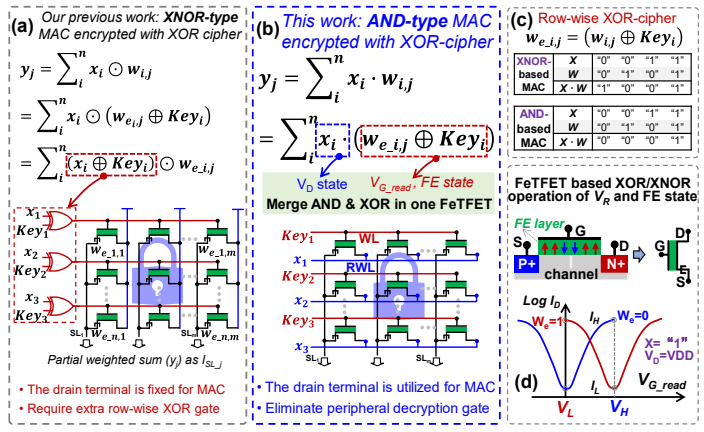


Fig. 2 (a) The FeTFET-based XNOR-type encrypted multiply-accumulate (eMAC) with extra XOR gate for decryption. (b) The proposed FeTFET-based AND-type eMAC in this work. (c) Truth table of XOR cipher and XNOR/AND type multiplication. (d) The FeTFET-based non-monotonic XOR/XNOR operation.

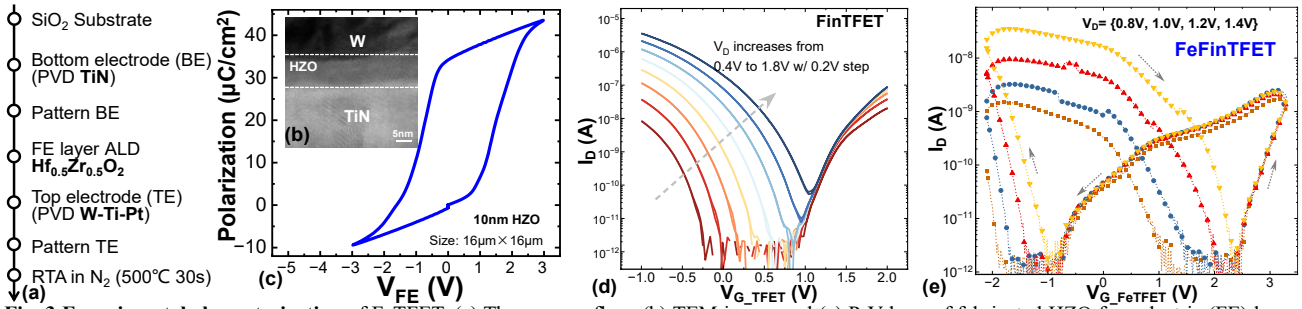


Fig. 3 Experimental characterization of FeTFET. (a) The process flow, (b) TEM image and (c) P-V loop of fabricated HZO ferroelectric (FE) layer. (d) The I_D-V_G characteristics of fabricated n-type TFET on a 14nm FinFET platform, and the drain voltage (V_D) modulated band-to-band tunneling (BTBT) behavior of FinTFET. (e) The I_D-V_G curves of FeFinTFET, which is constructed by connecting FE layer to the gate of FinTFET.

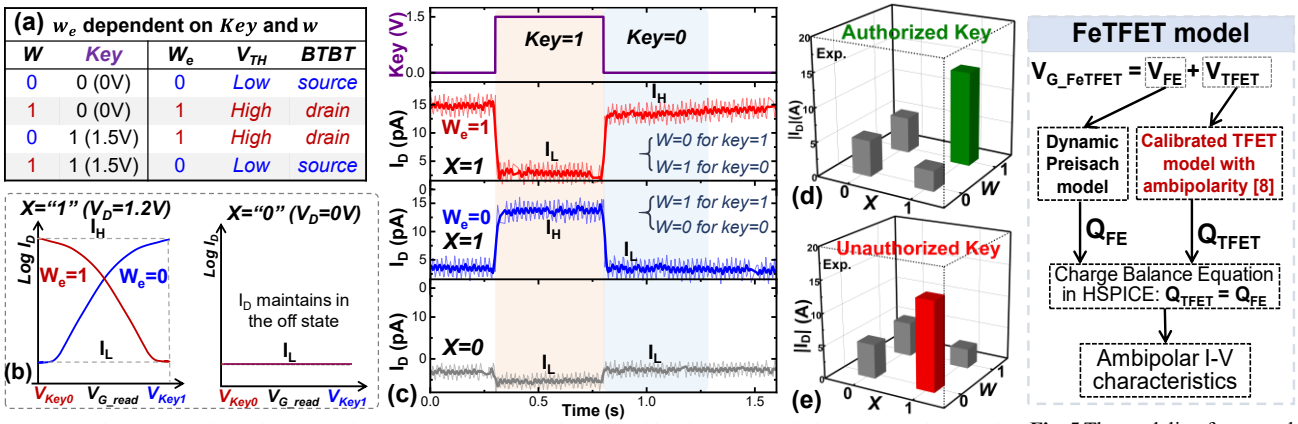


Fig. 4 (a) The encrypted weight (w_e) and FeFinTFET states are determined by the Key and plaintext weight (w). (b) The schematic diagram of FeFinTFET based XOR encrypted AND-type multiplication. (c-e) **Measured** XOR cipher encrypted AND-type MAC behavior of FeFinTFET-based nvCIM without explicit decryption.

Fig. 5 The modeling framework for FeTFET with ambipolar BTBT effect.

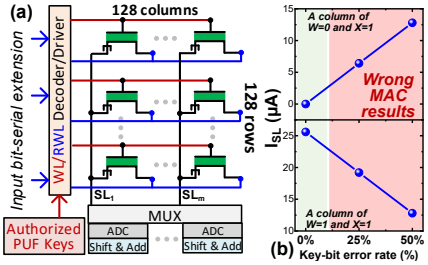


Fig. 6 (a) The schematic of FeTFET-based nvCIM with expandable precision of x and w. (b) The encrypted MAC operation in HSPICE.

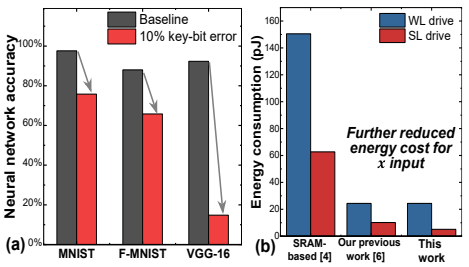


Fig. 7 (a) The encrypted neural networks based on the proposed secure nvCIM. (b) The comparison of energy consumption for MAC operation with decryption.

Table I Benchmark of proposed FeTFET-based secure nvCIM in this work with other reported solutions.

	ICCAD [4]	CICC [5]	Our previous work [6]	This work
Cell type	SRAM	RRAM	FeTFET	**FeTFET**
MAC type	AND		XNOR	**AND**
W_e cell size	6T	2T-2R	1T & row-wise XOR gate	**1T**
Energy (pJ)[a]	212.6	123.2	34.4	**29.4pJ**

(a) Energy cost for eMAC, calculated in HSPICE based on the encrypted array of the same size (128×128) with respectively proposed encrypted weight cell.

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